Chapter 8, Lecture 12

Marten Wegkamp

Sufficiency

Example We flip a biased coin n times and observe $x_1, \ldots, x_n \in \{0,1\} = \{H, T\}$. Our model is

$$\mathbb{P}\{X = 0\} = 1 - p \qquad \text{and} \qquad \mathbb{P}\{X = 1\} = p$$

for some unknown p. Rather than storing these n numbers, it suffices to record one single value, namely the number of successes (tails). We say that

$$T := \sum_{i=1}^{n} X_i = \#\text{successes}$$

is a sufficient statistic of p. It contains all there is to learn about p from the data.

It is of course also dangerous to throw away data – what if your model is wrong?

Definition $T = T(X_1, \ldots, X_n)$ is a sufficient statistic for θ if the conditional distribution of $\mathbf{X} = (X_1, \ldots, X_n)$ given $T = t$ does not dependent on the parameter θ, for any value t.
Hence, knowing T, we cannot learn more about θ from $\mathbf{X}$.

Example (continued). Note that

$$\mathbb{P}\{X = x\} = p^x (1-p)^{1-x}, \ x \in \{0,1\}$$

so that

$$\mathbb{P}\{\mathbf{X} = \mathbf{x}\} = p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i},$$

and

$$\mathbb{P}\{T = t\} = \binom{n}{t} p^t (1-p)^{n-t}$$

The conditional distribution of $\mathbf{X}$ given $T = t$ is independent of p. Consider two cases: $\sum_{i=1}^n x_i \neq t$ and $\sum_{i=1}^n x_i = t$.
1. If $\sum_{i=1}^n x_i \neq t$, then

$$\begin{aligned} \mathbb{P}\{\mathbf{X} = \mathbf{x} | T = t\} &= \frac{\mathbb{P}\{\mathbf{X} = \mathbf{x}, T = t\}}{\mathbb{P}\{T = t\}} \\ &= \frac{\mathbb{P}\{\emptyset\}}{\mathbb{P}\{T = t\}} \\ &= 0, \end{aligned}$$

which is clearly independent of t.
2. If $\sum_{i=1}^n x_i = t$, then

$$\begin{aligned} \mathbb{P}\{\mathbf{X} = \mathbf{x} | T = t\} &= \frac{\mathbb{P}\{\mathbf{X} = \mathbf{x}, T = t\}}{\mathbb{P}\{T = t\}} \\ &= \frac{\mathbb{P}\{\mathbf{X} = \mathbf{x}\}}{\mathbb{P}\{T = t\}} \\ &= \frac{p^{\sum_{i=1}^n x_i}(1-p)^{n-\sum_{i=1}^n x_i}}{\binom{n}{t} p^t (1-p)^{n-t}} \\ &= \frac{p^t (1-p)^{n-t}}{\binom{n}{t} p^t (1-p)^{n-t}} \\ &= \frac{1}{\binom{n}{t}}, \end{aligned}$$

which, again, does not depend on p.

Factorization theorem

A natural question is

How do we find a sufficient statistic?

Theorem *Let* $\mathbf{X} = (X_1, \ldots, X_n)$ *have joint pdf or pmf* $f(\mathbf{x}|\theta)$*. Then*

(1) $T = T(\mathbf{X})$ *is sufficient*

$\Longleftrightarrow$

(2) $f(x|\theta) = g(T(\mathbf{x}), \theta)h(\mathbf{x})$

That is, a necessary and sufficient condition for T to be sufficient is that the likelihood $\text{lik}(\theta)$ factorizes according to (2), a part $h(\mathbf{x})$ that depends on the data only, and a part $g(T(\mathbf{x}), \theta)$ that depends on θ and on the data *only* through $T(\mathbf{x})$.

EXAMPLE (continued).
The likelihood function factorizes as follows

$$\begin{aligned} \text{lik}(\theta) &= p^{\sum X_i}(1-p)^{n-\sum_{i=1}^n X_i} \\ &= p^T(1-p)^{n-T} \end{aligned}$$

and we can take $g(t, p) := p^t(1-p)^{n-t}$ and $h(\mathbf{x}) = 1$. We conclude that T is sufficient.

Proof: We only give the proof for the discrete case - the general case is a bit tricky. Hence the likelihood is simply

$$\text{lik}(\theta) = f(\mathbf{x}|\theta) = \mathbb{P}\{\mathbf{X} = \mathbf{x}\}.$$

Necessity of (2). If T is sufficient, that is, (1) holds, then by the definition of conditional probability, we find

$$\begin{aligned} \mathbb{P}\{\mathbf{X} = \mathbf{x}\} &= \mathbb{P}\{\mathbf{X} = \mathbf{x}|T = t\}\mathbb{P}\{T = t\} \\ &:= h(\mathbf{x})g(t, \theta) \end{aligned}$$

as $h(\mathbf{x}) := \mathbb{P}\{\mathbf{X} = \mathbf{x}|T = t\}$ does not depend on θ since $T = T(\mathbf{X})$ is sufficient, while $g(t, \theta) := \mathbb{P}_\theta\{T = t\}$ only depends on t and θ.

Sufficiency of (2). First, we observe that, using the factorization (2),

$$\begin{aligned}\mathbb{P}\{T=t\} &= \sum_{\mathbf{x}:\ T(\mathbf{x})=t} \mathbb{P}\{\mathbf{X}=\mathbf{x}\} \\ &= \sum_{\mathbf{x}:\ T(\mathbf{x})=t} g(t,\theta)h(\mathbf{x})\end{aligned}$$

We consider two cases:

1. If $T(\mathbf{x}) \neq t$, then

$$\mathbb{P}\{\mathbf{X}=\mathbf{x}|T(\mathbf{X})=t\}=0,$$

independent of θ.

2. If $T(\mathbf{x})=t$, then

$$\begin{aligned}\mathbb{P}\{\mathbf{X}=\mathbf{x}|T(\mathbf{X})=t\} &= \frac{\mathbb{P}\{\mathbf{X}=\mathbf{x}\}}{\mathbb{P}\{T(\mathbf{X})=t\}} \\ &= \frac{g(t,\theta)h(\mathbf{x})}{\sum_{\mathbf{y}:\ T(\mathbf{y})=t} g(t,\theta)h(\mathbf{y})} \\ &= \frac{h(\mathbf{x})}{\sum_{\mathbf{y}:\ T(\mathbf{y})=t} h(\mathbf{y})}\end{aligned}$$

which does not depend on θ. We conclude that T is sufficient. ■

We give a sleuth of examples:

(1) Normal(μ,σ^2). The joint density

$$f(\mathbf{x}|\mu,\sigma^2)=(2\pi)^{-n/2}\sigma^{-n}\exp\left(-\frac{\sum_{i=1}^n x_i^2-2\mu\sum_{i=1}^n x_i+n\mu^2}{2\sigma^2}\right)$$

is a function in $(\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2)$ only. In other words, we can take $h(\mathbf{x})=1$ and

$$g((\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2),(\mu,\sigma^2))=f(\mathbf{x}|\mu,\sigma^2)$$

By the factorization theorem, the pair $(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2)$ is sufficient for (μ,σ^2). Since any 1-1 function of a sufficient statistic is again sufficient, we also have

that $(\overline{X}_n, S^2)$ provide all the information the is to know about (μ, σ^2).

(2) Poisson(μ). Here we find

$$f(\mathbf{x}|\mu) = \left(\exp(-n\mu)\mu^{\sum_{i=1}^n x_i}\right)\left(\prod_{i=1}^n x_i!\right)^{-1} := g(\sum_{i=1}^n x_i, \mu)h(\mathbf{x})$$

so that $\sum_{i=1}^n X_i$ is sufficient.

(3) Exponential(λ).

$$f(\mathbf{x}|\lambda) = \lambda^n \exp(-\lambda \sum_{i=1}^n x_i)$$

so that $T = \sum_{i=1}^n X_i$ is sufficient.

(4) Uniform$(0, \theta)$. Recall

$$f(\mathbf{x}|\theta) = \theta^{-n} 1\{\theta \geq x_{(n)}\}$$

whence $T = X_{(n)} = \max(X_1, \ldots, X_n)$ is sufficient.

(5) Suppose X has a shifted (by θ) exponential(1) distribution, that is,

$$f(x|\theta) = \exp(-(x-\theta)),\ x \geq \theta.$$

Then

$$\begin{aligned} f(\mathbf{x}|\theta) &= \exp(-\sum_{i=1}^n (x_i - \theta)) 1\{x_{(1)} \geq \theta\} \\ &= \exp(-\sum_{i=1}^n x_i)\left(\exp(n\theta) 1\{x_{(1)} \geq \theta\}\right) \end{aligned}$$

Set

$$\begin{aligned} h(\mathbf{x}) &= \exp(-\sum_{i=1}^n x_i) \\ g(x_{(1)}, \theta) &= \exp(n\theta) 1\{x_{(1)} \geq \theta\} \end{aligned}$$

to conclude that $T = X_{(1)} = \min(X_1, \ldots, X_n)$ is a sufficient statistic for θ.